

STRAUBING, Germany

WMO: 107880

Lat: 48.8275N

Lon: 12.5597E

Elev: 1178

StdP: 14.08

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	12.6	3.0	6.7	7.6	8.8	9.0	13.5	25.6	45.4	22.4	43.3	3.1	90	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	85.3	67.3	81.9	66.1	79.1	65.1	69.3	81.2	67.8	78.7	66.3	76.4	5.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	97.8	74.3	63.9	92.9	72.1	62.6	88.7	70.6	34.1	81.7	32.8	78.8	31.6	76.6	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	15.7	13.2	0.6	90.7	6.6	2.8	-4.1	92.7	-8.0	94.3	-11.7	95.9	-16.5	97.9		
(5)			0.1	73.1	6.5	1.6	-4.6	74.2	-8.4	75.1	-12.1	76.0	-16.9	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	29.2	32.3	39.9	48.7	57.1	62.5	65.4	64.8	56.8	48.4	38.5	31.2	(6)
(7)		DBStd	14.64	8.71	8.94	7.24	7.09	6.28	6.30	5.32	5.19	5.63	6.58	6.56	7.86	(7)
(8)		HDD50	2627	647	495	318	106	12	1	0	0	7	109	349	583	(8)
(9)		HDD65	6409	1111	914	778	490	256	121	62	67	250	516	796	1048	(9)
(10)		CDD50	1892	0	0	5	66	231	377	478	460	213	59	3	0	(10)
(11)		CDD65	200	0	0	0	1	10	47	75	62	5	0	0	0	(11)
(12)		CDH74	2146	0	0	0	13	162	488	758	655	68	2	0	0	(12)
(13)		CDH80	503	0	0	0	1	21	107	208	163	3	0	0	0	(13)
(14)	Wind	WSAvg	5.6	5.8	6.3	6.7	6.1	5.6	5.1	5.2	4.8	5.0	5.1	5.6	5.9	(14)
(15)	Precipitation	PrecAvg	33.9	2.5	2.2	2.4	1.9	3.2	3.8	3.9	3.4	2.8	2.4	2.6	2.8	(15)
(16)		PrecMax	45.7	5.6	4.7	6.3	4.1	6.9	5.9	6.3	6.3	6.3	7.2	5.3	6.3	(16)
(17)		PrecMin	24.3	0.3	0.6	0.4	0.2	0.5	1.6	0.5	1.1	1.2	0.6	0.0	0.7	(17)
(18)		PrecStd	4.4	1.3	1.1	1.5	1.0	1.4	1.0	1.4	1.5	1.2	1.4	1.2	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	54.0	64.0	75.5	82.9	87.2	89.6	90.3	79.7	71.8	59.0	51.1	(19)
(20)			MCWB	47.6	47.8	50.9	57.3	64.5	69.2	69.8	68.1	65.4	60.7	52.8	47.5	(20)
(21)		2%	DB	48.7	50.6	59.4	71.0	78.7	83.1	85.5	84.2	75.9	65.2	53.6	46.6	(21)
(22)			MCWB	44.9	45.6	49.8	55.8	63.1	67.6	67.4	66.2	63.7	58.4	50.0	44.2	(22)
(23)		5%	DB	44.4	47.2	55.3	67.0	74.6	79.6	81.8	80.8	72.2	61.5	50.5	43.1	(23)
(24)			MCWB	41.6	43.4	47.6	54.1	60.8	66.0	66.6	65.3	61.9	56.3	47.9	41.5	(24)
(25)		10%	DB	40.7	43.8	51.2	62.8	70.6	76.2	78.1	77.5	68.5	58.3	47.6	40.3	(25)
(26)			MCWB	38.7	41.0	46.0	51.7	58.8	64.3	65.2	64.4	60.2	54.5	45.6	38.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.8	48.4	53.2	60.0	67.7	71.9	71.8	70.7	67.3	62.0	53.4	48.3	(27)
(28)			MCDWB	52.3	52.5	61.0	71.6	78.9	83.4	85.6	83.9	76.2	68.2	57.6	50.7	(28)
(29)		2%	WB	44.8	46.1	50.5	56.9	64.1	69.3	69.1	68.3	65.0	59.4	50.3	44.5	(29)
(30)			MCDWB	48.1	49.9	57.4	68.6	75.1	80.4	81.3	79.9	73.1	64.0	53.0	46.3	(30)
(31)		5%	WB	41.8	43.7	48.5	54.7	62.0	67.2	67.6	66.8	63.2	57.1	48.1	41.6	(31)
(32)			MCDWB	44.3	46.8	54.0	65.1	72.2	77.1	78.7	77.4	70.0	61.0	50.1	43.1	(32)
(33)		10%	WB	38.7	41.1	46.4	52.7	60.1	65.1	66.1	65.3	61.2	54.9	45.7	39.0	(33)
(34)			MCDWB	40.5	43.7	51.1	61.5	68.8	73.9	76.0	74.7	67.1	57.6	47.4	40.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	12.1	15.1	19.0	19.3	18.8	19.5	19.6	17.9	14.7	9.1	8.0	(35)
(36)			MCDBR	12.2	16.4	23.3	27.6	27.0	25.4	26.6	26.9	24.8	19.5	14.0	11.2	(36)
(37)			MCWBR	9.8	11.9	14.7	14.4	13.6	12.2	11.7	11.8	13.6	12.9	10.4	9.2	(37)
(38)		5% WB	MCDBR	12.0	15.6	20.8	25.1	24.2	23.6	23.4	23.6	21.3	17.7	13.2	10.8	(38)
(39)			MCWBR	9.9	11.8	13.8	13.8	12.8	11.9	10.9	11.2	12.4	12.2	10.3	9.2	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.302	0.324	0.362	0.395	0.402	0.405	0.413	0.407	0.380	0.353	0.327	0.299	(40)
(41)	taud		2.411	2.364	2.319	2.260	2.274	2.298	2.271	2.309	2.356	2.422	2.454	2.443	(41)	
(42)	Ebn at Noon		241	261	268	270	272	272	267	264	259	247	230	227	(42)	
(43)	Edh at Noon		21	27	34	40	41	40	41	37	32	26	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	305	595	930	1361	1587	1748	1681	1477	1053	652	325	232	(44)	
(45)		RadStd	53	82	118	174	188	151	161	151	140	74	37	28	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61
(47)	Regional (78 neighbors)	+0.76	N/A	N/A	N/A	+0.59	+0.67	N/A	N/A	+108	+41	

Nomenclature: See separate page